

Total No. of Questions : 8]

SEAT No. :

PE2750

[Total No. of Pages : 2

[6583]-334

T.E. (Honors) (Computer Engineering) (Virtual & Augmented Reality)

VIRTUAL REALITY

(2019 Pattern) (Semester - V) (310701)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8,
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Discuss the concept of Chaining Transformations, and give an example of its use in VR. **[10]**

b) Explain the role of human eye movements in VR and their implications for VR system design. **[8]**

OR

Q2) a) What role does transformation chaining play in VR, and how does it enhance a VR user's spatial awareness? **[10]**

b) How does understanding the physiology of human vision, specifically eye movements, influence the design of VR applications? **[8]**

Q3) a) Explain the concept of Depth Perception in VR and discuss two techniques used to create a sense of depth. **[10]**

b) How does the perception of motion contribute to realism in VR environments? Describe one challenge associated with motion perception in VR. **[8]**

OR

Q4) a) Describe how color perception is managed in VR and why it is important for user experience. **[10]**

b) What is Ray Tracing, and how does it differ from Rasterization in the context of VR rendering? **[8]**

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Q5) a) Describe the concept ofvection and its relevance in VR. What challenges does it present for VR design? **[8]**

b) How does the vestibular system contribute to motion perception, and what implications does it have for VR experiences? **[9]**

OR

Q6) a) Explain the importance of tracking 2D and 3D orientation in VR and provide an example of how it is implemented. **[8]**

b) What are the implications of mismatched motion in VR, and how do VR systems address this issue to enhance user comfort? **[9]**

Q7) a) Explain the concept of “motor programs” in the context of VR and how it relates to user interaction. **[8]**

b) Discuss the concept of “remapping” in motor programs during VR interaction. How does it improve user experience in VR? **[9]**

OR

Q8) a) Describe the role of locomotion in VR. What are the challenges in implementing realistic locomotion in virtual environments? **[8]**

b) What is the significance of social interaction in VR, and how do VR systems enable users to interact with others in virtual spaces? **[9]**

